Can a mix be safer than either fund?

Stock fund

20% volatility

Bond fund

10% volatility

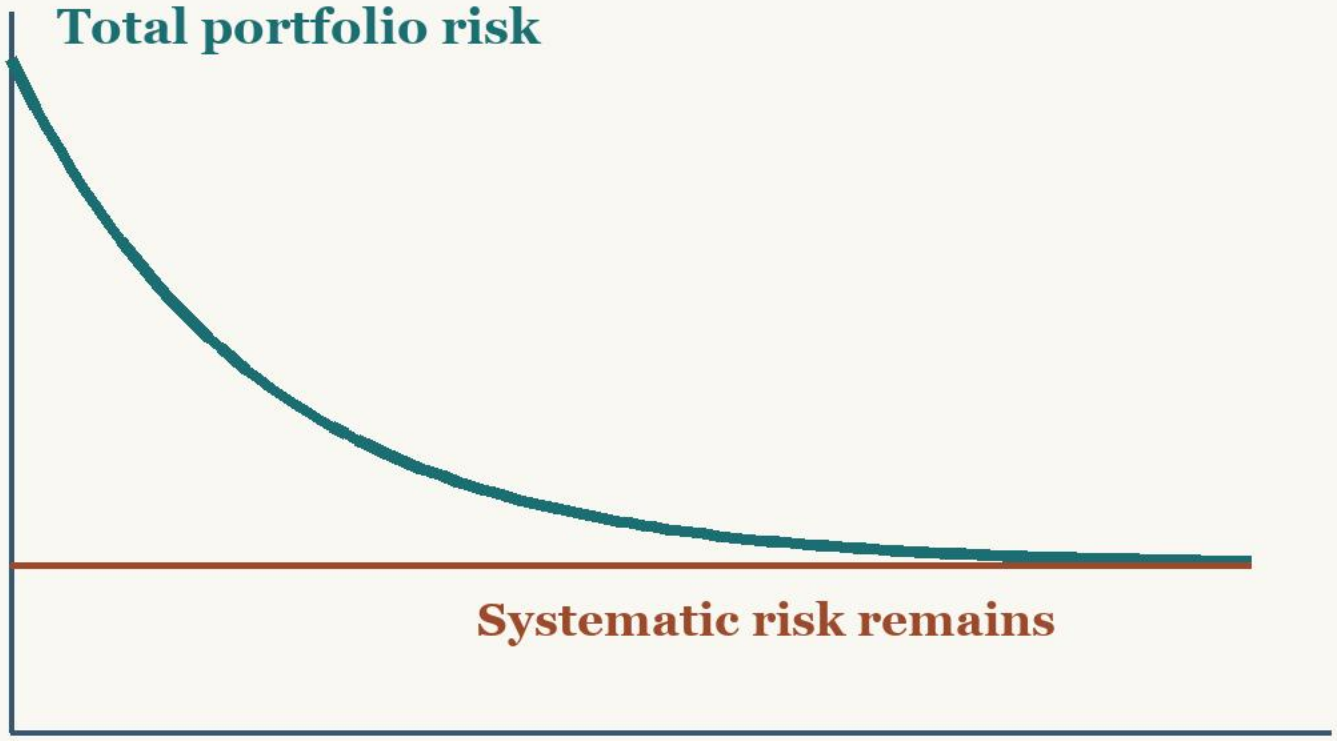


A stock–bond mix

Predict its volatility

Diversification depends on how returns move together.

Diversification removes firm-specific risk



Firm-specific

Recall, lawsuit, management change

Systematic

Recession, inflation, interest rates

A broad portfolio still faces recessions, inflation and rate changes.

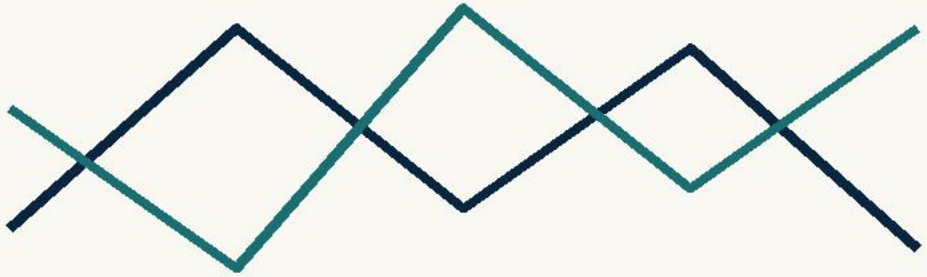
Different return patterns create the benefit

Same direction



Stock fund Second fund

Different patterns



Stock fund Second fund

Schematic return patterns across the same periods

More holdings help when their risks do not all move together.

Start with a distribution of possible returns

State	Probability	Stock return	Bond return
Expansion	20%	25%	4%
Stable growth	50%	10%	6%
Contraction	30%	−10%	2%

Probabilities total 100%. Each row pairs returns in the same economic state.

These are possible outcomes with known probabilities, not historical observations.

Expected return is the probability-weighted mean

$$\mu = E(R) = \Sigma\, p(i)R(i)$$

p(i) = state probability; R(i) = return in that state; μ = expected return

Weight each stock return

$0.20 \times 25\% + 0.50 \times 10\% + 0.30 \times (-10\%)$

Add the contributions

$5\% + 5\% - 3\% = 7.00\%$

Repeat for bonds

$0.20 \times 4\% + 0.50 \times 6\% + 0.30 \times 2\% = 4.40\%$

A 7% expected return is a long-run average across scenarios, not a promise.

Variance measures distance from the mean

$$\sigma^2 = \Sigma\, p(i)(R(i) - \mu)^2;\ \sigma = \sqrt{\sigma^2}$$

Stock state	Deviation from 7%	Weighted squared deviation
Expansion	$0.25 - 0.07 = 0.18$	$0.20 \times 0.18^2 = 0.00648$
Stable growth	$0.10 - 0.07 = 0.03$	$0.50 \times 0.03^2 = 0.00045$
Contraction	$-0.10 - 0.07 = -0.17$	$0.30 \times (-0.17)^2 = 0.00867$

$\sigma^2 = 0.01560$
$\sigma = 12.49\%$

Square deviations so gains and losses cannot cancel. SD returns to return units.

Build the statistics with cell references

F5: =(C5-C9)^2*B5

Cell / range	Stock calculation	Bond calculation
C9 / D9	=SUMPRODUCT(B5:B7,C5:C7)	=SUMPRODUCT(B5:B7,D5:D7)
F5 / G5	=(C5-C9)^2*B5	=(D5-D9)^2*B5
F9 / G9	=SUM(F5:F7)	=SUM(G5:G7)
B18 / C18	=SQRT(F9)	=SQRT(G9)

Fill F5:G5 through row 7. Results: stock μ 7%, σ 12.49%; bond μ 4.4%, σ 1.744%.

$ locks the mean while each state return and probability move down a row.

Covariance compares deviations in the same state

$$c = \mathrm{Cov}(S,B) = \Sigma\, p(i)(RS(i) - \mu S)(RB(i) - \mu B)$$

S = stock; B = bond; c = covariance; μ = each fund's own mean

Stock: Above	×	**Bond: Above**	→	**Positive**
Stock: Below	×	**Bond: Below**	→	**Positive**
Stock: Above	×	**Bond: Below**	→	**Negative**

Both above or both below their means add positive covariance.

Calculate the covariance one state at a time

State	p × stock deviation × bond deviation	Contribution
Expansion	0.20 × 0.18 × (−0.004)	−0.000144
Stable growth	0.50 × 0.03 × 0.016	0.000240
Contraction	0.30 × (−0.17) × (−0.024)	0.001224

c = 0.001320
Means: stock 7%; bond 4.4%

Weight the paired products by their probabilities, then add them.

Correlation puts covariance on a common scale

$$\rho = c / (\sigma S \times \sigma B)$$

ρ = correlation, from −1 to +1; σS and σB = fund standard deviations

$$0.001320 \div (0.124900 \times 0.017436)$$

$$\rho = +0.606$$



Positive correlation below +1 still permits diversification.

Your turn: compare two equally likely states

State	Probability	Stock return	Bond return
Strong economy	50%	20%	2%
Weak economy	50%	0%	6%

Find both means and SDs. Then calculate covariance and correlation. Predict the sign first.

Use each asset's own mean and match returns from the same state.

Opposite deviations create negative correlation

Measure	Stock fund	Bond fund
Mean	10%	4%
Variance	0.0100	0.0004
Standard deviation	10%	2%

$$c = 0.5(0.10)(-0.02) + 0.5(-0.10)(0.02) = -0.002$$
$$\rho = -0.002 / (0.10 \times 0.02) = -1$$

Here, one fund is above its mean exactly when the other is below.

Historical samples need sample statistics

Known scenario probabilities

Weight returns by probabilities

Variance = $\Sigma\, p(i)(R(i) - \mu)^2$

Use SUMPRODUCT and SQRT

Observed historical sample

Each observation is one data point

Sample variance divides by n − 1

Use VAR.S and STDEV.S

Q4 asks about a sample of 50 annual observations: use the .S functions.

One Excel function estimates sample volatility

Cell / range	Input or formula	Meaning / result
B5:B8	8%, −4%, 12%, 4%	Four hypothetical annual returns
B10	=AVERAGE(B5:B8)	5.00% mean
B11	=VAR.S(B5:B8)	0.00466667 variance
B12	=STDEV.S(B5:B8)	6.831% standard deviation

Squared deviations total 0.014. Divide by 4 − 1, then take the square root.

Use one full, numeric return range. Do not include years or mix asset columns.

Carry one stock–bond example through the model

Annual input	Stock fund	Bond fund
Expected return	12%	6%
Standard deviation	20%	10%
Variance	0.0400	0.0100

Correlation: 0.20

Risk-free rate: 3%

Base mix: 60% stocks + 40% bonds

Keep expected returns, standard deviations and correlation consistent.

Portfolio return weights the two fund means

$$E(R_P) = w_S\mu_S + w_B\mu_B;\ w_S + w_B = 1$$

w = portfolio share; μ = expected fund return; P = combined risky portfolio

60% stocks | **40% bonds**

0.60 × 12% = 7.20%

0.40 × 6% = 2.40%

7.20% + 2.40% = 9.60%

60% stocks and 40% bonds produce 9.60% expected return.

Portfolio variance has three terms

$$\sigma P^2 = wS^2\sigma S^2 + wB^2\sigma B^2 + 2wSwB\ c$$

$c = \rho\sigma S\sigma B = 0.20 \times 0.20 \times 0.10 = 0.004$

Stock term $0.60^2 \times 0.20^2 = 0.01440$

Bond term $0.40^2 \times 0.10^2 = 0.00160$

Co-movement term $2 \times 0.60 \times 0.40 \times 0.004 = 0.00192$

Add: 0.01440 + 0.00160 + 0.00192 = 0.01792

$\sigma P = \sqrt{0.01792} = 13.39\%$, below the 16% weighted average of fund SDs.

Your turn: increase the bond share

Given	Value
Stock expected return / SD	12% / 20%
Bond expected return / SD	6% / 10%
Correlation	0.20
New weights	20% stocks / 80% bonds

Calculate expected return, covariance, portfolio variance and SD. Can this mix be less volatile than bonds alone?

Show all three variance terms before taking the square root.

A small stock share can reduce bond-only risk

$$\sigma P^2 = 0.20^2(0.20^2) + 0.80^2(0.10^2) + 2(0.20)(0.80)(0.004)$$

Expected return

$0.20 \times 12\% + 0.80 \times 6\% = 7.20\%$

Portfolio variance

$0.00160 + 0.00640 + 0.00128 = 0.00928$

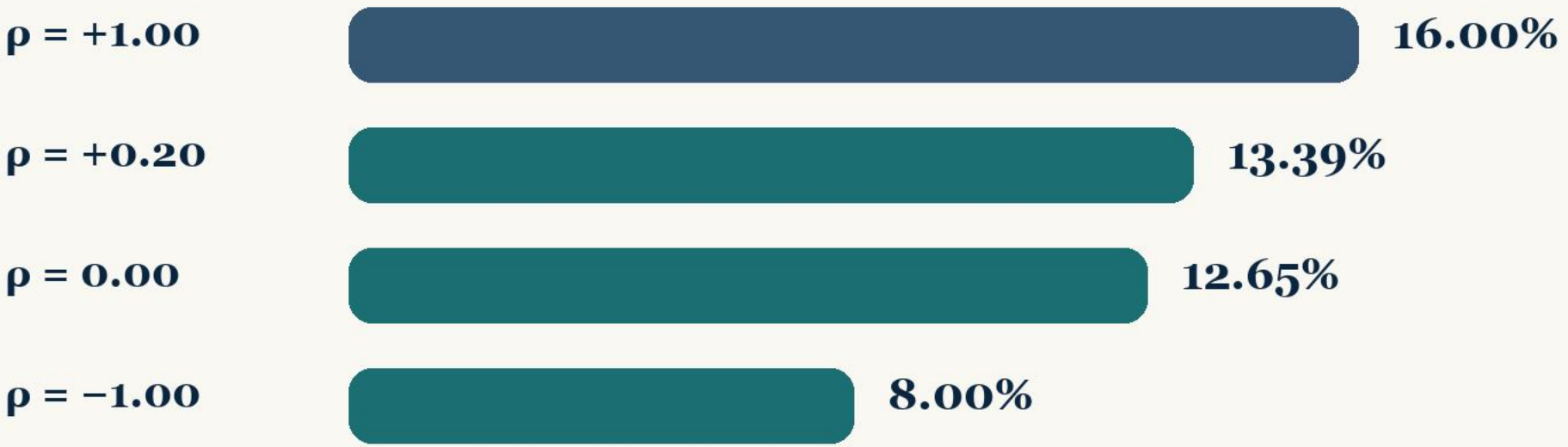
Portfolio standard deviation

$\sqrt{0.00928} = 9.633\% < 10\%$ bond-only SD

Imperfect co-movement can make the mix safer than either fund alone.

Lower correlation lowers risk at fixed weights

$$\sigma P^2 = wS^2\sigma S^2 + wB^2\sigma B^2 + 2wSwB\rho\sigma S\sigma B$$



ρ = −1 can produce zero risk only at special offsetting weights.

Expected return stays 9.60%; only the co-movement assumption changes.

Vary weights without moving fixed inputs

Cell / range	Input or formula	Purpose
B5:C5; B6:C6	12%, 6%; 20%, 10%	Means; SDs
A9; A17:A27	0.20; 0%, 10%, ..., 100%	Correlation; stock weights
B17	=A17*B5+(1-A17)*C5	Portfolio mean
C17	See formula below	Portfolio SD

C17: =SQRT((A17*B6)^2+((1-A17)*C6)^2
+2*A17*(1-A17)*A9*B6*C6)

Fill down: A17 changes to A18; the means, SDs and correlation stay locked.

Minimum variance finds the least volatile mix

$$w_{S,MV} = (\sigma_B^2 - c) / (\sigma_S^2 + \sigma_B^2 - 2c)$$

MV = minimum variance; $w_{B,MV} = 1 - w_{S,MV}$; c = covariance

Use the running-example risk inputs

$\sigma_S^2 = 0.04$; $\sigma_B^2 = 0.01$; $c = 0.004$

Calculate the stock share

$(0.01 - 0.004)/(0.04 + 0.01 - 0.008) = 14.29\%$

Calculate the resulting risk

85.71% bonds; expected return 6.857%; SD 9.562%

Minimum variance minimizes risk. It does not maximize the Sharpe ratio.

The efficient frontier is the upper boundary



Illustrative long-only two-fund opportunity set

Upper branch

Efficient risky portfolios

Leftmost point

Global minimum variance

Many-asset inputs

N means, N variances,
N(N − 1)/2 covariances

At the same risk, prefer the higher expected return under mean–variance assumptions.

Which candidate has the best reward per risk?

$$\text{Sharpe} = [E(RP) - Rf] / \sigma P$$

Candidate	Expected return	Standard deviation
100% stocks	12.00%	20.00%
60% stocks / 40% bonds	9.60%	13.3866%
100% bonds	6.00%	10.00%

Rf = 3%. Calculate and rank the three Sharpe ratios. Does the best candidate prove a global optimum?

Sharpe compares excess expected return per unit of total volatility.

60/40 wins the comparison, but is not optimal

100% stocks

0.450

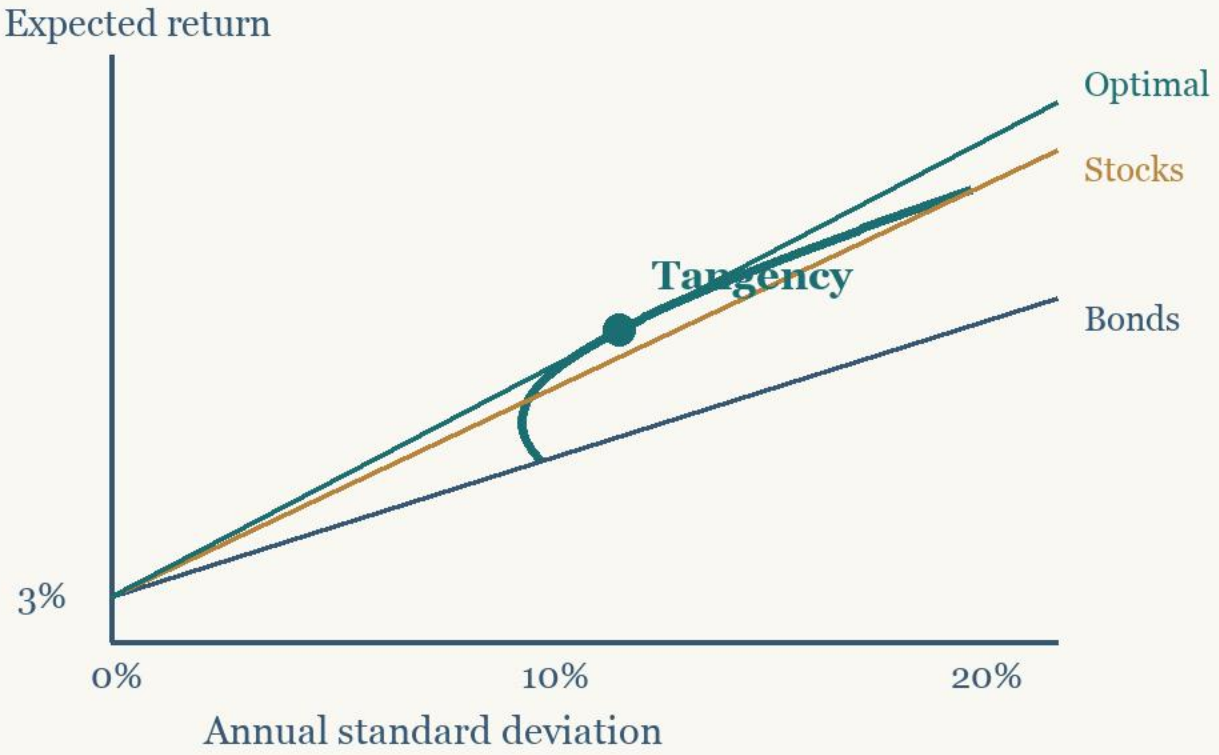
60/40 mix

0.493

100% bonds

0.300

Steeper CAL = higher Sharpe



Testing three candidates does not search every possible stock–bond mix.

Tangency weights maximize the Sharpe ratio

$$w_B^* = (e_B\sigma_S^2 - e_S\, c) / [e_B\sigma_S^2 + e_S\sigma_B^2 - (e_B + e_S)c]$$

$e_B = \mu_B - R_f$; $e_S = \mu_S - R_f$; $c = \rho\sigma_S\sigma_B$; $w_S^* = 1 - w_B^*$

Running example

$e_S = 12\% - 3\% = 9\%$
$e_B = 6\% - 3\% = 3\%$

$c = 0.004$
$\sigma_S^2 = 0.04$; $\sigma_B^2 = 0.01$

Bond numerator: $0.03 \times 0.04 - 0.09 \times 0.004 = 0.00084$

Denominator: 0.00162; bond weight = 51.85%

The optimal risky mix uses excess returns, variances and covariance.

Construct the tangency weights in Excel

Cell / range	Input or formula	Meaning
B5:C5	3%, 9%	Bond / stock excess returns
B6:C6; B9	10%, 20%; 0.004	Bond / stock SD; covariance
B17	Numerator ÷ denominator	Optimal bond weight
B18	=1-B17	Optimal stock weight

B17: =(B5*C6^2-C5*B9)/
(B5*C6^2+C5*B6^2-(B5+C5)*B9)

51.85% bonds + 48.15% stocks: expected return 8.889%, SD 11.815%.

Total returns must become excess returns first

Q6: excess returns supplied

Use the given excess returns directly

Bond inputs are in column B

Weights belong in B17:B18

Q7: total returns supplied

Subtract the Treasury-bill return

Stock excess return: B20

Stock weight: C20; bond: C21

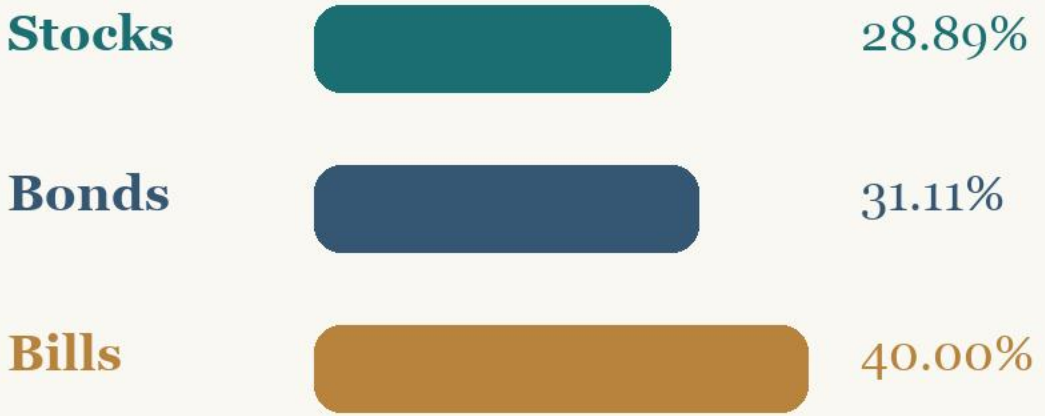
Use the same method, but read each question's labels and column order.

Choose the risky mix, then choose its size

Optimal risky mix

48.15% stocks
51.85% bonds

Choose y = 60% in the risky mix



$$E(RC) = yE(RP) + (1 - y)Rf;\ \sigma C = y\sigma P$$

CAL = bills + a risky portfolio. CML uses the market portfolio.

Risk preference changes the allocation to the risky mix and bills.

Diversification is powerful, but has limits

Estimation	Expected returns and correlations change; test assumptions.
Pooling and sharing	Pooling adds exposures. Sharing spreads a fixed risk among investors.
Time horizon	More years do not guarantee smaller cumulative losses.

Q1–Q2: measure / Q3: combine / Q4: sample / Q5: vary / Q6–Q7: optimize

Before trusting a portfolio, check inputs, formulas, constraints and economic meaning.